ARBA MINCH UNIVERSITY

ARBA MINCH INSTITUTE OF TECHNOLOGY
SCHOOL OF COMPUTING AND SOFTWARE ENGINEERING

INTERNSHIP REPORT

STORE MANAGEMENT SYSTEM

Prepared by: Gatluak James
Internship Host: Arba Minch University
Program: Computer Science
Internship Duration: [Insert Start Date – End Date]

Mentor: [Insert Mentor Name]
University Supervisor: [Insert Supervisor Name]
Date of Submission: [Insert Date]

DECLARATION

I, **Gatluak James**, declare that this internship report entitled **“Store Management System”** presents my internship work and learning experience at Arba Minch University. The report has been prepared for academic purposes in accordance with the internship report format provided by the Arba Minch University Internship and Externship Coordination Office.

Student Name: Gatluak James

Signature: _____

Date: _____

Mentor Approval: _____

University Supervisor Approval: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to God for giving me the strength, health, patience, and determination to complete my internship successfully.

I am grateful to Arba Minch University and Arba Minch Institute of Technology for providing the opportunity to gain practical experience and connect academic knowledge with an organizational working environment.

I sincerely appreciate the Internship and Externship Coordination Office, my internship mentor, university supervisor, staff members, colleagues, family, and friends for their guidance, encouragement, assistance, and support throughout the internship period.

EXECUTIVE SUMMARY

Internship provides students with an opportunity to connect classroom learning with practical work. This report presents the internship experience of Gatluak James at Arba Minch University, with the Store Management System serving as the main technical project considered during the internship.

The Store Management System is an inventory management application intended to automate the registration, tracking, management, and reporting of organizational materials. The project specification describes centralized inventory records, stock-in and stock-out transactions, returns, adjustments, employee and department management, supplier management, reporting, authentication, role-based access control, audit trails, and dashboard-based monitoring.

The internship strengthened practical understanding of software development, requirements analysis, system design, database-oriented thinking, inventory workflows, user access control, reporting, testing, documentation, teamwork, problem solving, and professional communication.

LIST OF FIGURES

Figure 1. Store Management System Dashboard — interface illustration

Figure 2. Material Management Interface — interface illustration

Figure 3. Inventory Transactions Interface — interface illustration

Figure 4. Role and Use-Case Overview

TABLE OF CONTENTS

Chapter One — Project Selection and Development

1.1 Project Title and Summary

1.2 Problem Statement

1.3 Justification

1.4 General and Specific Objectives

1.5 Scope

1.6 Methodology

1.7 Literature and Related Concepts

Chapter Two — Background of Arba Minch University

2.1 Brief History

2.2 Services and Academic Activities

2.3 Main Users and Stakeholders

2.4 Organizational Structure

2.5 Workflow

Chapter Three — Internship Experience and Specific Work

3.1 Reason for Selecting the Host Organization

3.2 Working Area

3.3 Work Process

3.4 Tasks Performed

3.5 Tools and Techniques

3.6 Challenges

3.7 Measures and Solutions

3.8 Results and Discussion

3.9 Recommendations

Chapter Four — Benefits Gained

Chapter Five — Conclusion and Recommendations

References / Source Materials

CHAPTER ONE — PROJECT SELECTION AND DEVELOPMENT

1.1 Project Title and Short Summary

Project Title: Store Management System

The Store Management System is a comprehensive inventory management application designed to automate the registration, tracking, and management of materials stored in an organization. According to the project specification, the system replaces manual record keeping with a centralized digital platform that provides current information about materials and inventory transactions.

The system covers material registration, inventory transactions, employees and departments, suppliers, reports, user management, authentication, access control, and audit logging. It also supports monitoring of total, issued, and remaining quantities.

1.2 Problem Statement

Manual or fragmented inventory processes can make it difficult to maintain consistent records of materials, stock movements, recipients, suppliers, approvals, and transaction history. Such processes may require repeated data entry and can make timely reporting more difficult.

The project therefore focuses on a centralized system that records inventory information and transactions in a structured manner. The system is intended to improve visibility, accountability, reporting, and control of organizational materials.

1.3 Justification

A digital inventory system can organize material information and provide a consistent record of stock received, stock issued, returns, adjustments, and related transactions. It can also help management obtain reports and monitor stock levels.

1.4 Objectives

- Automate registration and management of store materials.
- Track stock received, issued, transferred, returned, and adjusted.
- Maintain accurate total, issued, and remaining stock quantities.
- Record employees or departments receiving materials.
- Improve transparency, accountability, and inventory accuracy.
- Generate reports for management decision-making.
- Reduce material loss, duplication, and stock shortages.

1.5 Scope

The project specification covers material registration and categorization, stock receiving and issuing, returns, adjustments, transaction history, employee and department management, suppliers, user authentication, role-based access, reports, low-stock notifications, and audit logs.

1.6 Methodology

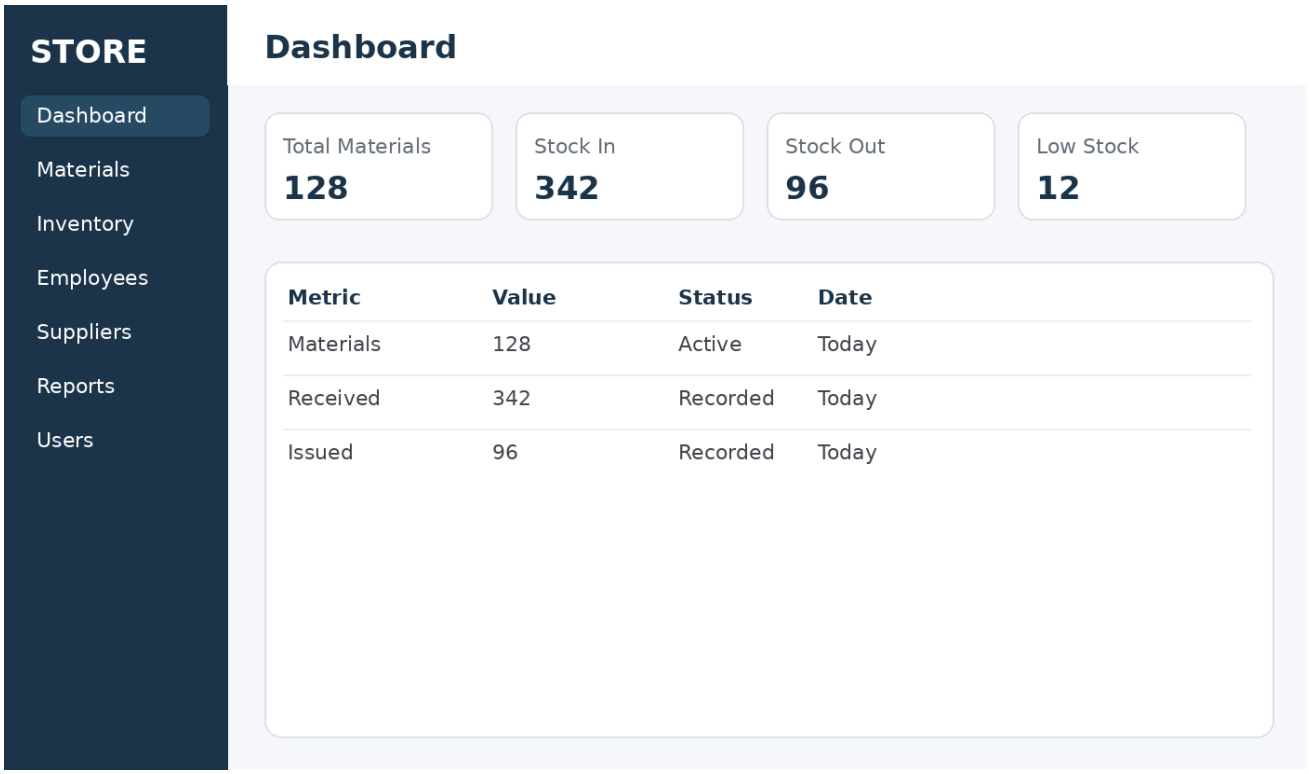
The work was approached as a practical software-development and systems-analysis exercise. The major activities include understanding the inventory problem, identifying users and use cases, defining

required data, organizing functional modules, considering transaction workflows, developing or documenting system interfaces, testing expected operations, and preparing technical documentation.

1.7 Literature and Related Concepts

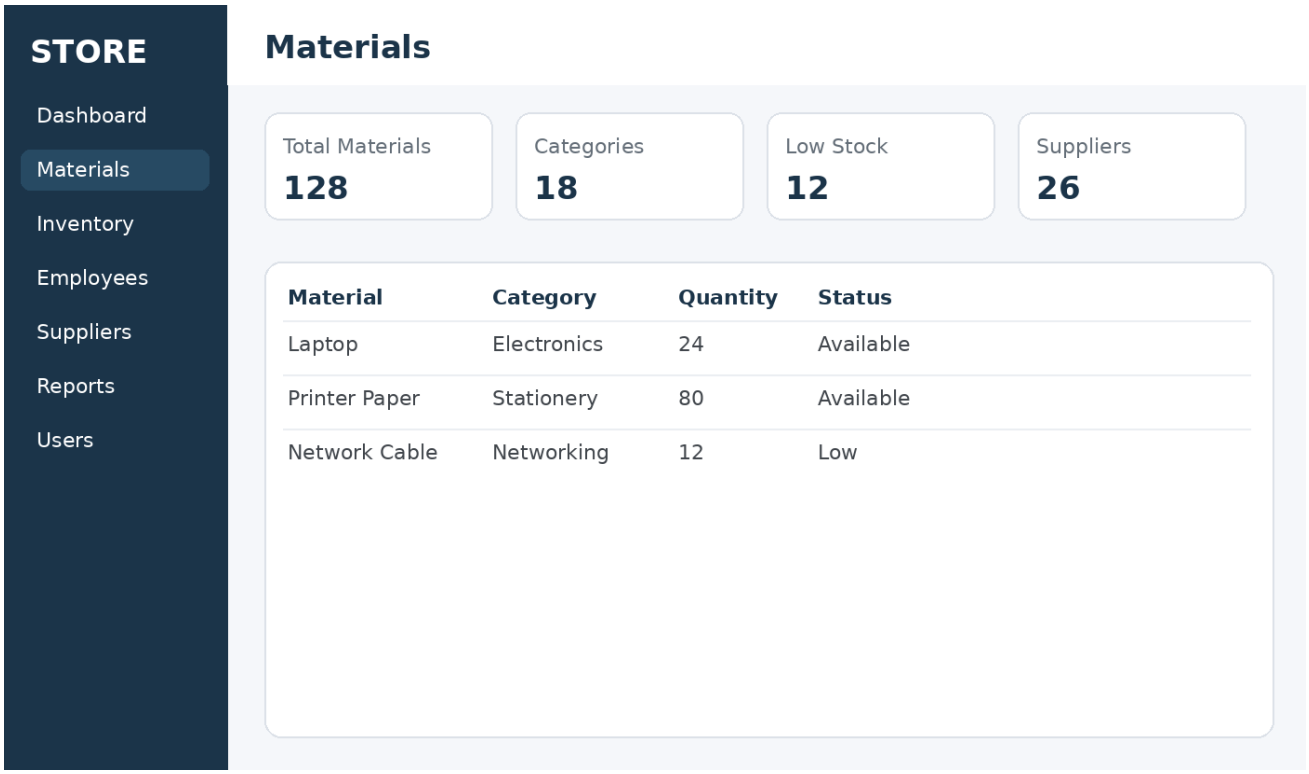
The project is related to common concepts in inventory management systems, database systems, information systems, software engineering, authentication, authorization, transaction processing, reporting, and audit trails. Inventory systems generally require reliable records of stock movement and clear identification of the people or departments responsible for transactions.

FIGURE 1 — SYSTEM INTERFACE



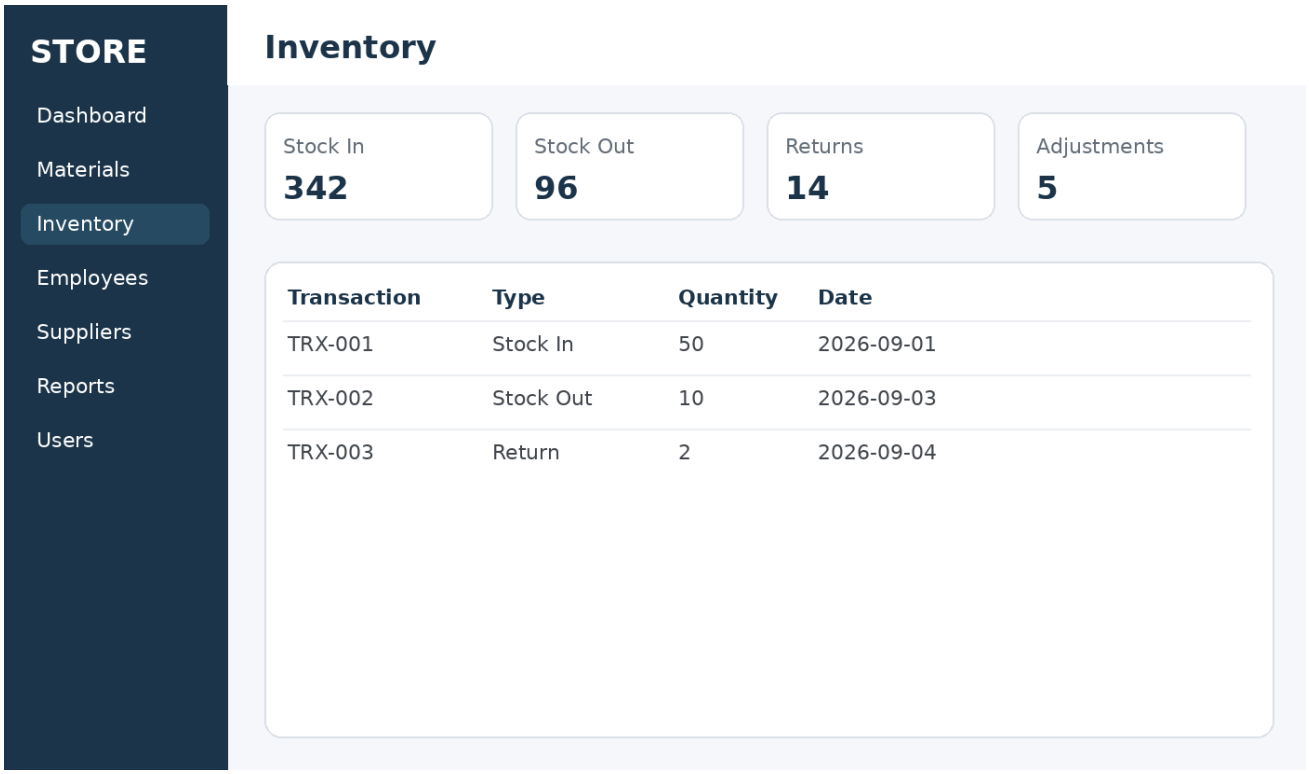
Interface illustration created from the uploaded Store Management System specification. Replace this illustration with the actual application screenshot before final submission if required.

FIGURE 2 — SYSTEM INTERFACE



Interface illustration created from the uploaded Store Management System specification. Replace this illustration with the actual application screenshot before final submission if required.

FIGURE 3 — SYSTEM INTERFACE



Interface illustration created from the uploaded Store Management System specification. Replace this illustration with the actual application screenshot before final submission if required.

CHAPTER TWO — BACKGROUND OF THE INTERNSHIP HOST ORGANIZATION

2.1 Brief History

Arba Minch University (AMU) is a public university in South-West Ethiopia. The university states that it was initially founded as the Arba Minch Water Technology Institute and was officially inaugurated as a full-fledged university in June 2004. It offers undergraduate and graduate programs through institutes, colleges, schools, and a campus including Arba Minch Institute of Technology. ■cite■turn0search7■

2.2 Main Services and Academic Activities

Arba Minch University provides higher education, research, community engagement, and related academic and institutional services. Its official website identifies multiple academic units, including Arba Minch Institute of Technology, and describes the university as a national research university. ■cite■turn0search7■

The university's official website also provides institutional information, announcements, systems, downloads, and communication services for students, staff, and other stakeholders. ■cite■turn0search2■

2.3 Main Users and Stakeholders

The main stakeholders of a university environment include students, academic staff, administrative staff, researchers, management, service units, and external partners. For an inventory/store management application, relevant operational users include administrators, store managers, storekeepers, employees or departments receiving materials, and reporting or auditing personnel.

2.4 Organizational Structure

The university consists of multiple institutes, colleges, schools, campuses, and administrative units. The project documentation specifically models roles within the store-management environment, including Administrator, Store Manager, and Storekeeper.

2.5 Workflow

The store workflow begins with registering materials and their categories, receiving stock, maintaining stock balances, issuing materials to employees or departments, recording returns and adjustments, and producing reports. User roles determine which operations can be performed.

CHAPTER THREE — INTERNSHIP EXPERIENCE AND SPECIFIC WORK

3.1 Why Arba Minch University Was Selected

Arba Minch University provided a relevant academic and institutional environment for gaining practical experience in information technology and software-related activities. Working in a university environment also provided exposure to real organizational processes and the importance of information management.

3.2 Working Area

The internship work described in this report focused on software and information-system activities related to the Store Management System. The project required understanding how inventory information is registered, processed, controlled, and reported.

3.3 Workflow of the Working Area

The workflow can be summarized as: user authentication → material registration → stock receiving → inventory update → material request or issue → approval and transaction recording → returns or adjustments → reporting and audit review.

3.4 Work Tasks Performed

- Analyzing the requirements of an organizational inventory system.
- Identifying material-management functions.
- Understanding stock-in and stock-out workflows.
- Organizing employee and department information.
- Considering supplier information and supplied materials.
- Defining reports required for inventory monitoring.
- Understanding user roles and access control.
- Considering transaction history and audit trails.
- Reviewing dashboard information and inventory statistics.
- Preparing project documentation and presentation materials.

3.5 Tools and Techniques

The work involved software-development concepts including requirements analysis, database-oriented data organization, user-interface design, role-based access control, transaction processing, reporting, system documentation, and testing. The project specification identifies support for dashboard statistics, search and filtering, audit trails, low-stock notifications, and export of reports to PDF and Excel.

3.6 Major Challenges

- Maintaining consistent and accurate inventory records.
- Representing different stock transactions without creating conflicting quantities.
- Designing access rules for different user roles.
- Keeping transaction history understandable and traceable.
- Providing useful reports from stored transaction data.

- Designing interfaces that are simple enough for routine store operations.

3.7 Measures Taken to Address Challenges

- Organizing the system into functional modules.
- Separating material information from transaction information.
- Using role-based access concepts.
- Recording transaction details such as issuer, approver, recipient, purpose, and remarks.
- Maintaining transaction history and audit logs.
- Using dashboards and reports to improve visibility.
- Applying search and filtering to make information easier to locate.

3.8 Results and Discussion

The project specification describes a centralized platform capable of recording materials and their categories, tracking stock movement, calculating remaining quantities, managing users and roles, and producing inventory-related reports. It also identifies benefits such as improved accountability, faster transactions, real-time stock visibility, and more comprehensive reporting.

3.9 Recommendations

- Maintain regular backups of inventory data.
- Apply strong authentication and role-based permissions.
- Keep an audit trail for important transactions.
- Perform regular stock reconciliation between physical stock and system records.
- Train users before deploying the system in routine operations.
- Review reports periodically to identify low-stock or unusual transactions.

USE CASE AND USER ROLES

Administrator

The project specification assigns the Administrator high-level control over system operations. Functions include user management, system settings, reports, dashboard access, audit-log monitoring, material management, inventory management, supplier management, employee/department management, and authentication.

Store Manager

The Store Manager is responsible for operational control and approval processes. The documented functions include approving or issuing materials, managing materials and inventory, viewing dashboards, generating reports, tracking stock levels, viewing transaction history, and managing employees/departments and suppliers.

Storekeeper

The Storekeeper handles day-to-day inventory operations including stock receiving, stock issuing, returns, adjustments, inventory updates, stock-level tracking, transaction history, basic material-record management, and authentication.

CHAPTER FOUR — BENEFITS GAINED FROM THE INTERNSHIP

4.1 Improvement of Practical Skills

The internship provided practical exposure to analyzing organizational problems and translating them into software requirements. It strengthened understanding of how inventory processes can be represented through system modules, data fields, user roles, and transaction workflows.

4.2 Upgrade of Theoretical Knowledge

Classroom concepts related to databases, software engineering, information systems, user authentication, system analysis, and application design became easier to understand when considered in the context of an actual organizational problem.

4.3 Industrial Problem-Solving Capability

The project required identifying problems associated with inventory control and considering structured solutions. This strengthened analytical thinking and the ability to divide a larger problem into manageable functional modules.

4.4 Teamwork Skills

Working in an organizational environment helped develop the ability to communicate requirements, receive feedback, discuss problems, and coordinate work with other people.

4.5 Leadership Skills

The internship encouraged responsibility, planning, task organization, decision-making, and ownership of assigned work.

4.6 Work Ethics and Professional Conduct

The internship strengthened understanding of punctuality, responsibility, respect, confidentiality, proper communication, documentation, and accountability in a professional environment.

4.7 Entrepreneurship Skills

The project demonstrated how a real organizational problem can become an opportunity for a practical software solution. It encouraged thinking about efficiency, user needs, service quality, and sustainable digital solutions.

4.8 Interpersonal Communication

The internship improved the ability to explain technical ideas in understandable language, ask relevant questions, receive feedback, and communicate professionally with colleagues and supervisors.

CHAPTER FIVE — GENERAL CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The internship at Arba Minch University provided valuable practical exposure to software and information-system development. The Store Management System project demonstrated how an organizational inventory problem can be analyzed and represented as a structured digital solution.

The project includes material registration, categorization, stock receiving, issuing, returns, adjustments, employee and department management, supplier management, reporting, authentication, role-based access control, dashboards, notifications, and audit trails. These functions address important requirements for organized inventory management.

The internship also helped connect theoretical knowledge with practical problem solving. It improved technical understanding, analytical ability, documentation skills, professional communication, teamwork, and awareness of workplace responsibilities.

5.2 General Recommendations

- Continue improving digital inventory management and reduce unnecessary manual record keeping.
- Provide adequate user training before operational deployment.
- Maintain reliable backup and recovery procedures.
- Review user permissions regularly.
- Conduct periodic physical-versus-system stock reconciliation.
- Use reports and audit logs for accountability and management review.
- Continue improving the user interface based on feedback from actual users.
- Consider barcode or QR-code support where it is practical.

5.3 Final Reflection

The internship was an important bridge between academic study and practical work. It showed that successful software development is not only about writing code; it also requires understanding people, processes, data, security, usability, documentation, and organizational goals.

REFERENCES AND SOURCE MATERIALS

1. Arba Minch University Internship and Externship Coordination Office. *Internship Report Writing Format*. User-provided guideline.

2. Store Management System project specification. User-provided internship/project document.

3. Arba Minch University. Official institutional information and profile. ■cite■turn0search7■turn0search12■

4. Arba Minch University official website and institutional announcements. ■cite■turn0search2■

NOTE FOR FINAL SUBMISSION

The uploaded Store Management System document contains the project specification but does not contain embedded application screenshots. The figures in this report are therefore interface illustrations based on that specification, not claimed photographs of the actual running application. Replace them with your real screenshots before final academic submission if your department requires actual system screenshots.

The official Arba Minch University website provides the university's institutional information and an official profile containing its emblem/logo. ■cite■turn0search12■turn0search7■